

PUBLIC VERSION

**UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.**

**Before the Honorable Clark S. Cheney
Chief Administrative Law Judge**

In The Matter Of

**CERTAIN SEMICONDUCTOR DEVICES,
PRODUCTS CONTAINING SAME, AND
COMPONENTS THEREOF**

Investigation No. 337-TA-1500

DECLARATION OF DR. FRANK ERNST REGARDING CLAIM CONSTRUCTION

I. INTRODUCTION

1. I, Frank Ernst, Ph.D., have been asked by counsel to submit this Opening Claim Construction Declaration on behalf of Tower Semiconductor Ltd., Tower Partners Semiconductor, Ltd., Tower Semiconductor Italy S.r.l., Tower US Holdings, Inc., Tower Semiconductor San Antonio, Inc., Tower Semiconductor NPB Holdings, Inc., Tower Semiconductor Newport Beach, Inc., or Newport Fab LLC (collectively, “Respondents”). I understand GlobalFoundries U.S. Inc. (“Complainant”) has alleged that Respondents infringe claims 1, 5-9, 10, 11, 12, and 15-19 (“Asserted Claims”) of U.S. Patent 8,330,235 (the “’235 patent” or “Asserted Patent”).

2. I have been asked by Respondents’ counsel to assess certain terms of the ’235 patent from the perspective of a person of ordinary skill in the art (“POSA”). To assess the claims of the ’235 patent, I have considered the teachings of the specification, prosecution history, scientific literature, the parties’ contentions and discovery responses, and any other documents and/or cited in this declaration. In addition to these materials, I may consider additional documents and information in forming any supplemental opinions.

3. I am being compensated for my time in connection with this declaration at my standard consulting rate of \$500 per hour. My compensation does not depend on the outcome of or the content of my testimony in the current proceedings and I have no financial interest in Respondents or Complainant.

II. QUALIFICATIONS

4. Below, I summarize my background, qualifications, and experience relevant to the issues raised in the current case. A copy of my curriculum vitae (“CV”), which includes a full description of my background qualifications, is attached to this declaration as Exhibit E and which

I hereby incorporate by reference. Within the past 5 years, I was retained as an expert in an unrelated investigation by Respondents Caihong Display Devices Co., Ltd. and Xianyang CaiHong Optoelectronics Technology Co., Ltd. (“Caihong”) entitled “Certain Glass Substrates for Liquid Crystal Displays, Products Containing the Same, and Methods for Manufacturing the Same II; Inv. No. 337-TA-1441.” I testified as an expert witness in deposition and trial. To the best of my knowledge and recollection, I have not testified in any other deposition or at trial within the past 5 years.

5. I am currently a (full) professor in the Department Chair of Materials Science and Engineering in the School of Engineering at Case Western Reserve University. For 9 years, until October 2024, I have been leading this department as department chair.

6. I have more than 40 years of experience in materials science and engineering, physics, mathematics, and computation. My scientific work contributed to several areas of material science, including ceramics. I have substantial expertise in microcharacterization of materials, including atomic-resolution imaging, diffractometry, and spectrometry using state-of-the-art sophisticated methodology and major instrumentation. I have served as the director of a microscopy center at Case Western Reserve University for about 10 years.

7. A major part of my scientific work involves semiconductor heterostructures, particularly based on Si and Ge–Si. The list of materials my group and I researched and published on includes Si, Ge–Si, III–V compound and chalcopyrite semiconductors, oxide ceramics, perovskite ceramics, carbides, nitrides, and metallic glasses. I also have expertise with the methodology and application of many sophisticated experimental methods that are in use to analyze the composition, microstructure, and atomistic structure of semiconductors and related interfaces, *e.g.* TEM, XEDS, EELS, SEM, FIB, AFM, XPS, AES, XRD, ToF-SIMS. At Case

Western Reserve University, I have been teaching three graduate-student-level courses on these techniques. My research group has advanced the frontier of research across a great variety of materials classes and phenomena, focusing on processing–composition–crystal/micro-structure–property relationships and with semiconductor materials playing a significant role. I am knowledgeable on the solid-state physics and materials science of semiconductors in general, including structure, electrical, and optical properties, and their interactions and interfaces with other materials that are relevant to integrated circuits, e.g. conducting and dielectric materials. The title of my habilitation thesis (the habilitation is the highest academic degree in Germany, similar to doctor of science in the UK) translates to “Structure of Heterointerfaces.” For 12 years, I was a senior scientist, leading a research group, in the (translated) “Department of Electron Microscopy and Inner Interfaces” at the Max-Planck-Institut für Metallforschung in Stuttgart, Germany. I collaborated with world renowned semiconductor epitaxy groups on research projects sponsored by the German government, such as a major project aiming to develop Si–Ge heterostructures for quantum-effect electronic devices. I am also an expert in thermodynamics of materials, diffusion, and diffusion theory. For example, my scientific work includes experimental (Boltzman–Matano) analysis and computer simulation of concentration-dependent diffusion. At Case Western Reserve University, I have been teaching an undergraduate course on “Thermodynamic Stability and Rate Processes” for several years.

III. APPLICABLE LEGAL PRINCIPLES

8. I am not an attorney. I am not providing any legal opinions. I have been informed by counsel of legal standards which I have applied in forming my opinions.

9. Counsel has informed me that claim construction is a question of law with underlying fact. As such, I understand that the Chief Administrative Law Judge decides questions of claim construction. I further understand from counsel that the purpose of claim construction

is to determine the meaning and scope of the asserted claims from the standpoint of a person of ordinary skill in the art (“POSA”).

10. I understand that claim terms should be read in the context of the specification and prosecution history. I understand that the specification and the prosecution history are considered parts of the “intrinsic record.”

11. I further understand that if the meaning of a claim term remains unclear in view of the intrinsic record, the court may consider “extrinsic evidence.” I understand that extrinsic evidence includes, among other things, expert testimony.

12. I have been informed that claim construction is evaluated from the perspective of a POSA at the time of the alleged invention. I understand from counsel that the following things may be considered to determine who qualifies as a POSA: (1) the types of problems encountered in the art, (2) prior art solutions to those problems, (3) the sophistication of the technology, (4) the rapidity with which innovations are made, and (5) the education level of active workers in the field. In the next section of this declaration, I have provided my opinion regarding the level of skill of a POSA.

13. I have further been informed that a patent claim is invalid as indefinite if the claim, when read in light of the specification and the prosecution history, fails to inform, with reasonable certainty, those skilled in the art about the scope of the invention. I understand that a claim term does not become “reasonably certain” simply because a skilled artisan may be able to articulate a definition for that term. I understand from counsel that showing that a term is indefinite does not require a showing that a term could not be understood at all. Instead, it requires demonstrating that a POSA does not have “reasonable certainty” regarding the meaning of the term.

IV. PERSON OF ORDINARY SKILL IN THE ART (“POSA”)

14. I understand that an evaluation of the claims of the Asserted Patents should be undertaken from the perspective of a person of ordinary skill in the art (“POSA”) as of the earliest priority date of the Asserted Patent. For purposes of this claim construction declaration, counsel has asked me to assume that the relevant date is February 5, 2009, which is the filing date of the earliest application that the ’235 patent claims priority to. I have further been informed by Counsel that Complainant has not identified an earlier priority date for the ’235 patent.

15. The relevant technical field for the asserted patents is the design and fabrication of semiconductor devices, including the formation of nickel platinum silicides. In my opinion, based on the legal principles I discussed above, my experience in and knowledge of the field and those who work and study in it, and the materials I have considered in preparing this declaration, a POSA in this technical field on February 5, 2009, would have had at least a Bachelor of Science in electrical engineering, materials science, materials engineering, materials science and engineering, chemical engineering, or similar majors with 2-3 years of experience related to the design and fabrication of semiconductor devices, including contact formation. More education could substitute for experience or vice versa.

16. My opinions in this declaration would not change if the POSA were determined to have somewhat more or less or slightly different education and/or experience than what I have summarized above. My knowledge and skill exceed that of a POSA and my opinions herein reflect, and are made from the perspective of, a POSA as of February 5, 2009.

V. MATERIALS CONSIDERED

17. In arriving at my opinions in this declaration, I have considered the asserted patents, their file histories, the parties’ proposed claim constructions, and all other information and documents cited or discussed in this declaration, including the exhibits below.

Exhibit No.	Description
-------------	-------------

PUBLIC VERSION

N/A	Complaint, <i>Certain Semiconductor Devices, Products Containing Same, and Components Thereof</i> , Investigation No. 337-TA-1500 (“Complaint”), and accompanying exhibits and appendices
N/A	Certified copy of U.S. Patent No. 8,330,235 (Exhibit 1 to Complaint) (“’235 patent”)
Appx. A	Certified copy of Prosecution History of the U.S. Patent No. 8,330,235 (Appendix A to Complaint) (GF-ITC-00004310 – GF-ITC-00004484)
E	Dr. Frank Ernst Curriculum Vitae
17	Praneet Adusumilli et al, <i>Silicide-phase evolution and platinum redistribution during silicidation of Ni_{0.95}Pt_{0.05}/Si(100) specimens</i> , 112 J. Appl. Phys. 064307 (2012) (“Adusumilli”)
18	Complainant GlobalFoundries U.S. Inc.’s Second Supplemental Objections and Responses to Respondents’ First Set of Interrogatories (Nos. 1-73)
19	Complainant’s Infringement Contentions, Ex. A1
20	Complainant’s Domestic Industry Contentions, Ex. B1

VI. THE ’235 PATENT

A. Asserted Claims of the ’235 Patent

18. Asserted claims 1, 5-9, 10, 11, 12, 15-19 of the ’235 patent recite as follows:

1. A semiconductor device comprising:

a silicon substrate,

a transistor, comprising source/drain regions and a gate electrode, on the substrate;

and

a nickel silicide layer on the gate electrode and source/drain regions,

wherein the nickel silicide layer comprises platinum having a composition gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direction away from the source/drain regions and the gate electrode, from between about 8 wt. % and about 12 wt. % at the silicon interface.

5. The semiconductor device according to claim 1, wherein the nickel silicide layer has a thickness of about 100 Å to about 150 Å.

6. The semiconductor device according to claim 1, wherein the silicon substrate comprises a strained silicon germanium (Si—Ge) layer on a silicon layer.

7. The semiconductor device according to claim 1, further comprising a stress liner on the nickel silicide layer.
8. The semiconductor device according to claim 7, further comprising an oxide or oxynitride layer on the stress liner.
9. The semiconductor device according to claim 8, wherein the stress liner comprises stressed silicon nitride.
10. A semiconductor device comprising:
 - a silicon substrate;
 - a P-channel transistor and an N-channel transistor, each comprising source/drain regions and a gate electrode, on the silicon substrate; and
 - a first nickel silicide layer on the gate electrode and source/drain regions of the P-channel transistor; and
 - a second nickel silicide layer on the gate electrode and source/drain regions of the N-channel transistor,wherein each of the first and second nickel silicide layers comprises platinum having a concentration gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direction away from the P-channel and N-channel transistors, from between about 8 wt. % and about 12 wt. % at the silicon interface.
11. The semiconductor device according to claim 10, further comprising:
 - a first stress liner on the P-channel transistor; and
 - a second stress liner on the N-channel transistor.

12. The semiconductor device according to claim 11, wherein the first stress liner exhibits high compressive stress and the second stress liner exhibits high tensile stress.
15. The semiconductor device according to claim 12, wherein each nickel silicide layer has a thickness of about 100 Å to about 220 Å.
16. The semiconductor device according to claim 15, wherein each nickel silicide layer has a thickness of about 100 Å to about 150 Å.
17. The semiconductor device according to claim 12, wherein the silicon substrate comprises a strained silicon germanium (Si—Ge) layer on a silicon layer.
18. The semiconductor device according to claim 12, further comprising an oxide or oxynitride layer on the first and second stress liners.
19. The semiconductor device according to claim 18, wherein the first and second stress liners comprise stressed silicon nitride.

B. Summary of the Specification of the '235 Patent

19. The '235 patent issued on December 11, 2012 and claims priority to U.S. Application No. 12/366,378 which was filed on February 5, 2009. The '235 patent is entitled "Method to Reduce MOL Damage on NiSi."

20. The '235 patent's "disclosure relates to semiconductor devices with improved nickel silicide layers" and states that its "disclosure is particularly applicable in fabricating transistors with improved device performance by avoiding nickel silicide degradation" during device processing. '235 patent, 1:11-15. The '235 patent discloses a two-step nickel deposition process for forming a silicide, which include "forming a first layer of nickel having a first amount of platinum and forming, on the first layer of nickel, a second layer of nickel having a second amount of platinum, the second weight percent of platinum being greater than the first weight

percent” and “anneal[ing those layers to] to form a nickel silicide layer having the platinum composition gradient increasing in platinum toward the upper surface.” ’235 patent, Abstract.

21. The specification explains that “[c]onventional semiconductor devices, such as the one shown in FIG. 1, include a refractory metal deposited over all exposed upper surface features of a silicon substrate **101**, preferably by means of physical vapor deposition (PVD).” ’235 patent, 1:19-22. “The most common refractory metal employed is nickel, which forms a very low resistivity silicide with silicon and does not consume as much Si as other silicides commonly employed.” ’235 patent, 1:22-25. “NiSi has a great resistance to etchants which are used to etch silicon oxide, a common insulating material used in semiconductor devices.” ’235 patent, 1:25-28.

22. The specification further explains that:

Generally, in forming the MOS transistor, the refractory metal is deposited after etching the gate electrode **103** and after forming the source/drain junction **105**, as illustrated in FIG. 1. After deposition, the refractory metal layer blankets the top surface of the gate electrode **103**, sidewall spacers **107**, oxide liners **109**, and source/drain regions **105**. As a result of thermal processing, e.g., a rapid thermal annealing (RTA) process, performed in an inert or reducing atmosphere, the refractory metal reacts with underlying silicon to form electrically conductive silicide layer portions **111** on the top surface of the polysilicon gate electrode **103** and on the exposed surfaces of the substrate where source and drain regions **105** are formed. Unreacted portions of the refractory metal layer, e.g., on the silicon oxide, nitride or oxynitride sidewall spacers **107** and the silicon oxide liner **109** are then removed, as by a wet etching process selective to the metal silicide portions **111**.

’235 patent, 1:41-57.

23. Figure 1 of the ’235 patent is reproduced below.

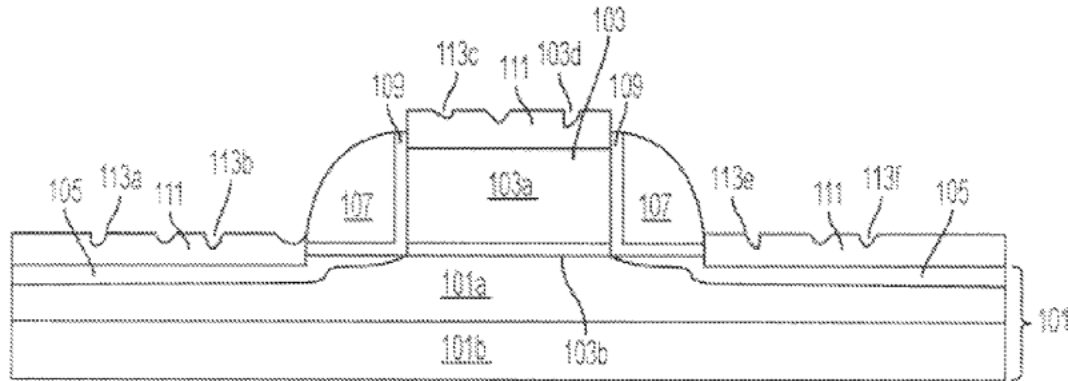


FIG. 1
(PRIOR ART)

'235 patent, Fig. 1.

24. The '235 patent further discloses that “[s]ubstrates based on ‘strained silicon’ have attracted interest as a semiconductor material which provides increased speeds of electron and hole flow therethrough, thereby permitting fabrication of semiconductor devices with higher operating speeds, enhanced performance characteristics, and lower power consumption.” '235 patent, 1:58-63. Stress liners may also be added to further improve carrier mobility. '235 patent, 2:28-49.

25. When stress liners are used, “reactive ion etching” is necessary. The '235 patent explains that:

Because the stress liners differ for P-channel and N-channel transistors, the stressed dielectric layers are typically patterned by reactive ion etching, which involves physically bombarding the surfaces to subtractively remove material. This etching process involves an “overetch” in order to clear the residue left behind. It is during this overetch process that the underlying NiSi layers are damaged, causing degradation of the NiSi. The degradation can be morphological, wherein mouse bites or missing NiSi is observed, as illustrated in FIG. 1 at **113 a-113 f**, or electrical, wherein the resistivity is increased. Either or both types of degradation

ultimately result in poor device parameters such as higher sheet resistance and external resistance.

'235 patent, 2:50-62.

26. To solve this purported issue, the '235 patent discloses:

[F]orming two layers of nickel on a silicon substrate and a silicon gate electrode, the first layer having a relatively low concentration of platinum and the second having a relatively high concentration of platinum, and annealing the two layers to form a single layer of NiSi. During annealing, platinum diffuses to form a composition gradient with the amount of platinum increasing in a direction away from the substrate and gate electrode. It was found that such a concentration gradient of platinum effectively shields the NiSi layer from physical damage during subsequent processing, e.g., etching, while concomitantly maintaining the integrity of the NiSi layer.

'235 patent, 5:12-23.

27. The '235 patent further discloses:

Adverting to FIG. 3, a first layer of nickel 301 is deposited over the source/drain regions 213b, the spacers 211, the oxide liner 209, and the gate electrode 205. First layer of nickel 301 may be formed, for example, with a platinum content of about 8 wt. % to about 12 wt. %, such as about 10 wt. %. The first layer of nickel can be formed by sputtering using a target comprising Ni and 10 wt. % Pt. Over the first layer of nickel 301, a second layer of nickel 303 is deposited. The second layer of nickel 303 may be formed, for example, with a platinum content of about 30 wt. % to about 70 wt. %, such as about 50 wt. %. The second layer of nickel can be sputtered using a target of Ni containing an amount of Pt within the range of 30 to 100%. The first nickel layer 301 and second nickel layer 303 may be formed to a combined thickness of about 45 Å to about 100 Å using a conventional deposition tool, for example on an AMAT Endura sputtering system, at a power of 1500 W to 2500 W, e.g., 2000 W, and using an Argon flow of 20 sccm to 40 sccm, e.g., 2000 sccm, or an Argon and Hydrogen mixture with a flow of 3 sccm to 10 sccm, e.g., 5 sccm. The first layer of nickel 301 may be formed at a thickness of about 60% to about 80% of the combined thickness, such as about 70% of the combined thickness.

A capping layer 305 may be formed over the second nickel layer 303 to prevent oxidation of the underlying metal. Capping layer 305 may be titanium nitride.

RTA and etch processes are performed to form a NiSi layer 401 over the source/drain regions 213 b and the gate electrode 205, as illustrated in FIG. 4. The resulting NiSi layer 401 has a composition gradient of platinum from about 8 wt. % to about 12 wt. %, such as about 10 wt. %, at the silicon interface 403 to about 30 wt. % to about 70 wt. %, such as about 30 wt. % to about 50 wt. %, at the opposite surface. NiSi layer 401 may be formed at a thickness of about 100 Å to about 220 Å, such as at a thickness of about 100 Å to about 150 Å.

'235 patent, 6:45-7:12; *see also* '235 patent, 8:41-52.

28. The '235 patent further discloses a two-step annealing process that “reduces the time at which annealing occurs at a high temperature, thereby reducing the roughness of resulting interfaces between the NiSi and the underlying Si layer.” '235 patent, 7:43-46.

VII. CLAIM CONSTRUCTION TERM AND THE PARTIES' PROPOSED CONSTRUCTION

29. I understand that a single term has been identified for the '235 patent:

Disputed Claim Term	Complainant's Proposed Construction	Respondents' Proposed Construction
“from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide” '235 patent, claims 1, 10	Plain and ordinary meaning; not indefinite.	Indefinite

30. Below I provide my opinions regarding the above identified term. I reserve the right to address other issues as they arise.

VIII. CLAIM CONSTRUCTION

A. “from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide”

Disputed Claim Term	Complainant's Proposed Construction	Respondents' Proposed Construction
“from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide” '235 patent, claims 1, 10	Plain and ordinary meaning; not indefinite.	Indefinite

31. In my opinion, the asserted claims of the '235 patent are indefinite because the term “from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide” fails to inform a POSA about the scope of the alleged inventions with reasonable certainty, when read in light of the patent’s specifications and prosecution histories.

32. It would confuse a POSA that the '235 patent talks about a “nickel silicide *surface*,” but a “silicon *interface*.” From the perspective of a POSA, this terminology is undefined and inconsistent.

1. The Specification

33. In my opinion, the specification lacks guidance regarding the *location* of the “nickel silicide surface at a silicon interface” or the “opposite surface of the nickel silicide.” The specification identifies element 403 in Fig. 4 as the “silicon interface.” In my opinion this figure provides no guidance to a POSA because it provides no information regarding the actual position of the interface or the structure of the interface at an atomistic level. For example, what is the mole (or mass) fraction of each element on either side of the “interface” to identify it as an “interface”?

34. After finite time at finite temperature, and definitely during the annealing described in the '235 patent, solid–solid interfaces generally undergo *intermixing* by diffusion. The “sharp” interface that two solids A and B may make at the moment of first getting in contact

loses its identity because atoms from solid A diffuse into B and atoms from solid B diffuse into A. Generally, the transport rate differs between atom species and depends on the local environment – atomistic structure and chemical potentials of all species in the system. Therefore, after any finite amount of time has passed after initial contact at finite temperature, the originally sharp interface will be *diffuse*. That means it will no longer be straight forward to assign an exact location to the interface, unless by an arbitrary criterion. For the concrete case of Ni–Pt in contact with Si, Ni and Pt will both diffuse into Si. The exact location of “a silicon interface,” as conceived in the ’235 patent, is no longer meaningful unless determined by an arbitrary criterion. The ’235 patent, however, does not provide any such criterion.

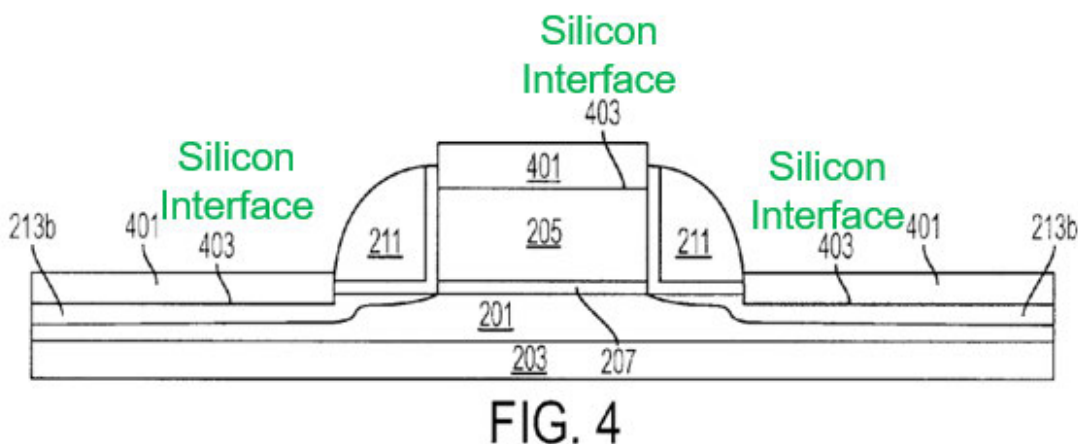


FIG. 4

'235 patent, Fig. 4 (annotated).

35. In fact, the specification only uses the term “silicon interface” once in the specification:

RTA and etch processes are performed to form a NiSi layer 401 over the source/drain regions 213b and the gate electrode 205, as illustrated in FIG. 4. The resulting NiSi layer 401 has a composition

gradient of platinum from about 8 wt. % to about 12 wt. %, such as about 10 wt. %, at the silicon interface 403 to about 30 wt. % to about 70 wt. %, such as about 30 wt. % to about 50 wt. %, at the opposite surface. NiSi layer 401 may be formed at a thickness of about 100 Å to about 220 Å, such as at a thickness of about 100 Å to about 150 Å.

'235 patent, 7:4-12 (emphasis added).

36. Elsewhere in the specification, the term “interface” is used:

The two-step anneal may be conducted by first heating at a temperature of about 280° C. to about 350° C., for example about 300° C. to about 320° C., e.g., 300° C., during which metal rich silicide, Ni₂Si, is formed. During the second step, the temperature may be raised to about 360° C. to about 450° C., such as about 380° C. to about 420° C., e.g., 420° C. During the second annealing step, the metal rich silicide is converted to monosilicide. The total time for the two step annealing can be the same as for the single step anneal process, for example 5 seconds for the first step and 30 seconds for the second step. The two-step anneal therefore reduces the time at which annealing occurs at a high temperature, thereby reducing the rough interfaces between the NiSi and the underlying Si layer.

'235 patent, 7:33-46 (emphasis added).

37. In my opinion, a POSA would not have been reasonably certain as to where the “nickel silicide surface at a silicon interface” is located based on the language of the specification. The prosecution history also fails to provide any guidance. In particular, the '235 patent does not provide any guidance as to how the location of the “nickel silicide surface at a silicon interface” is to be *measured*.

38. A number of acceptable methods of “microcharacterization” are in use for this purpose, but they yield *different* results. For example, researchers have used XEDS (X-ray Energy-Dispersive Spectrometry) in conjunction with SEM (Scanning Electron Microscopy) to measure profiles $X_i[z]$ or $W_i[z]$ of the atom fraction X or the mass fraction W of elements $i, i = 1 \dots N$, and determine surface/interface positions from the characteristics of the profiles.

However, depending on the exact imaging and measurement parameters, this method only provides limited spatial resolution and involves significant uncertainties regarding the levels of individual elements, especially for light elements. The correlation between the abundance of a given element in the specimen and the intensity of its characteristic X-rays measured by a detector is *complicated*. Quantification of the specimen composition requires precise knowledge on the element-specific *sensitivity* of the X-ray detection setup. However, the sensitivity depends on many factors, including the element-specific yield of X-rays, their absorption in the specimen and the detector window, and secondary X-ray emission via fluorescence. Absorption and fluorescence depend on the composition – which the experiment tries to determine and which, therefore, is unknown. Moreover, these measurements are typically carried out on cross-sections of bulk specimens with the incident electron beam normal to the polished surface. In this setting, X-rays typically escape from a spherical volume with a diameter of 1 μm below the surface. In addition to this rather poor intrinsic resolution limit, real surfaces/interfaces can be *curved* in three dimensions. Therefore, the results are not only impacted by the rather limited spatial resolution, but also by averaging over specimen depths in which the surfaces/interfaces reside in different lateral positions and exhibit different inclinations against the direction of the incident electron beam.

39. Another approach is to perform compositional analysis by XEDS and/or EELS (Electron Energy-Loss Spectrometry) in conjunction with TEM (Transmission Electron Microscopy). As the specimen required for TEM is a foil, typically less than 0.1 μm thick, this provides higher spatial resolution and suffers less from projection effects of laterally displaced or inclined surfaces/interfaces. However, the sample volume is correspondingly small and therefore TEM suffers from lesser general validity of the observations. As shown below, a major drawback

of determining composition profiles by XEDS-TEM linescans is the presence of significant random *fluctuations* in composition profiles. And, similar to XEDS-SEM, the quantification of XEDS-TEM data suffers from uncertainties related to limited knowledge on the sensitivity of the individual X-ray detector system to X-rays from specific elements. Obtaining representative data of high reliability will require a large number of measurements and specimens, which are laborious and costly to make.

40. A superior experimental approach is provided by atom-probe tomography. This is a modern technique that analyzes a needle-shaped specimen prepared from the material under study. It can determine the atom number about every other atom in the needle and its position in the needle with atomistic 3D resolution. Sophisticated techniques of statistical atom-position-data analysis have been developed in the context of this experimental method in years after the '235 patent was filed. These methods can properly address curvature of interfaces and provide "3D composition profiles," known as proxigrams. Instead of averaging different lateral positions, proxigrams reveal the true compositional change as a function of distance normal to the surfaces/interfaces.

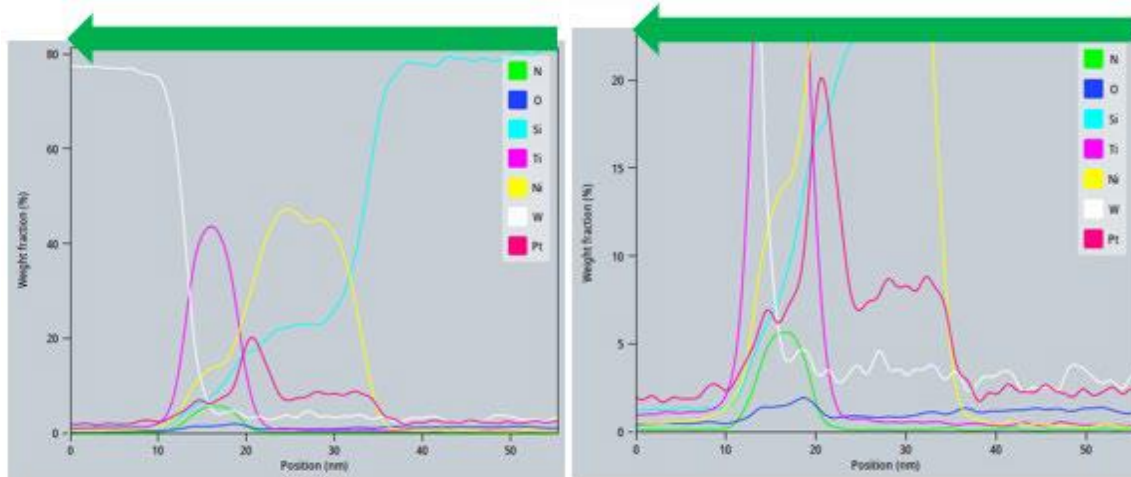
41. In view of the above, it is my opinion that a POSA would not have had reasonable certainty as to the location of the "nickel silicide surface at a silicon interface." For example, claims 1 and 10 of the '235 patent require a "platinum increasing in amount from between about 8 wt. % and about 12 wt. % *at the silicon interface*." See '235 patent, claims 1, 10 (emphasis added).

42. Firstly, determining the "amount" of Pt "at" the "silicon interface" is not reasonably certain to a POSA: Specifying the mole or mass fraction of an element requires a three-dimensional *volume*. The mole or mass fraction of an element within that volume follows

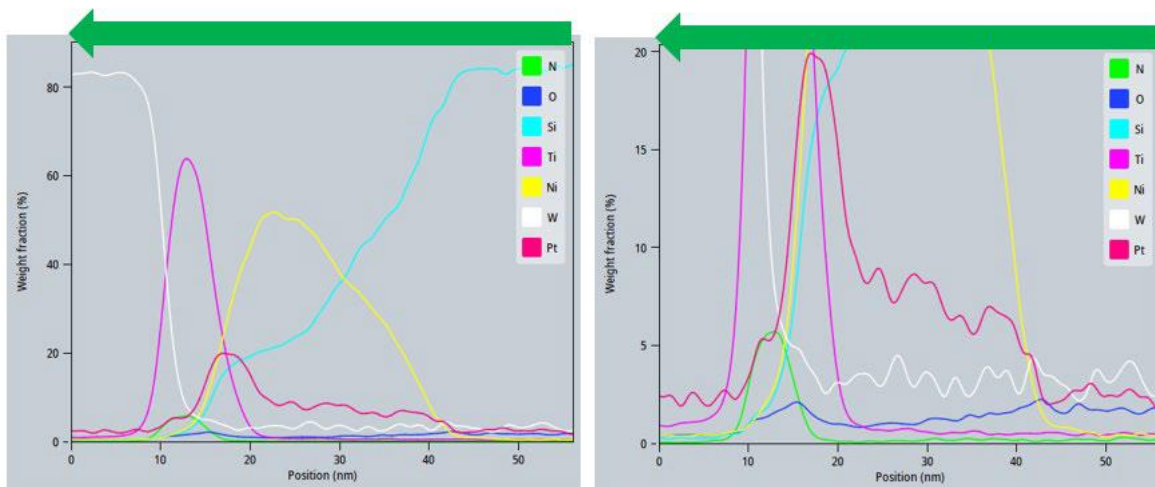
from counting the numbers of different atom species in the volume and dividing the results by the total. An interface, however, is a plane. It does not have a three-dimensional volume. Thus, it cannot contain any atoms. The number of atoms of any species contained in a certain area of interface and their total will all be equal to 0 (zero). Mole or mass fractions of atoms exactly at an interface will be expressions of the form $0/0$, which are mathematically indefinite. Therefore, a POSA would not understand how to measure the mass fraction of Pt at the “Si interface” (or the “amount” of Pt, in the language of the '235 patent).

43. Secondly, in my opinion, a POSA would not have had reasonable certainty for the scope of the claims given the ambiguity regarding the *location* of the Si interface. For example, a POSA would not have any explanation from the specification regarding where the Si interface is located nor how to measure that, and would therefore be unable to meaningfully measure the Pt level (mole fraction or mass fraction). As a result, a POSA would not have had any reasonable certainty for judging whether a level of Pt at a particular location in the silicide is within the scope of the '235 patent claims.

44. In my opinion, Complainant's contentions to date further demonstrate the lack of reasonable certainty provided by the claims, the specification, and the prosecution history. For example, Complainant failed to identify a “silicon interface” and “opposite surface” of the silicide in the composition profiles in the figure below, which I have taken from Complainant's infringement contentions:

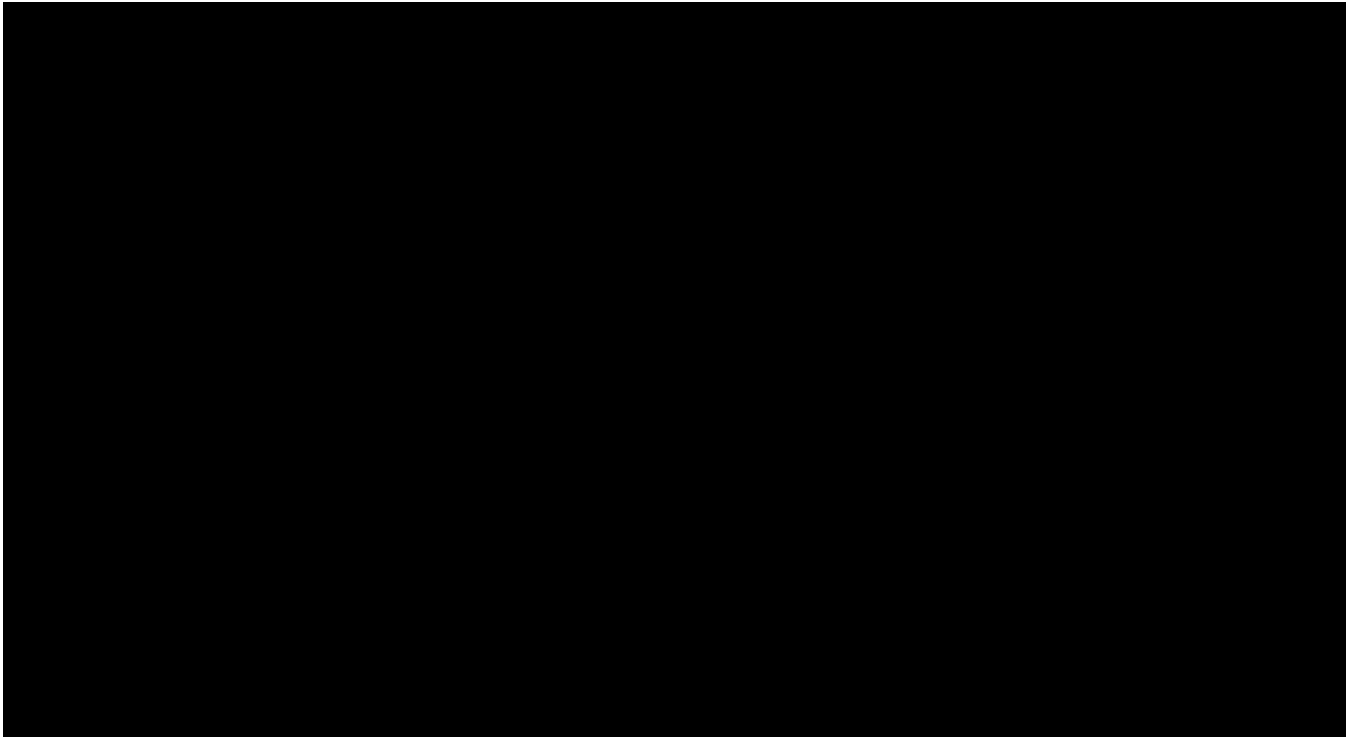


Ex. 19, Complainant's Infringement Contentions, Ex. A1 at 8.



Id. at 9-10.

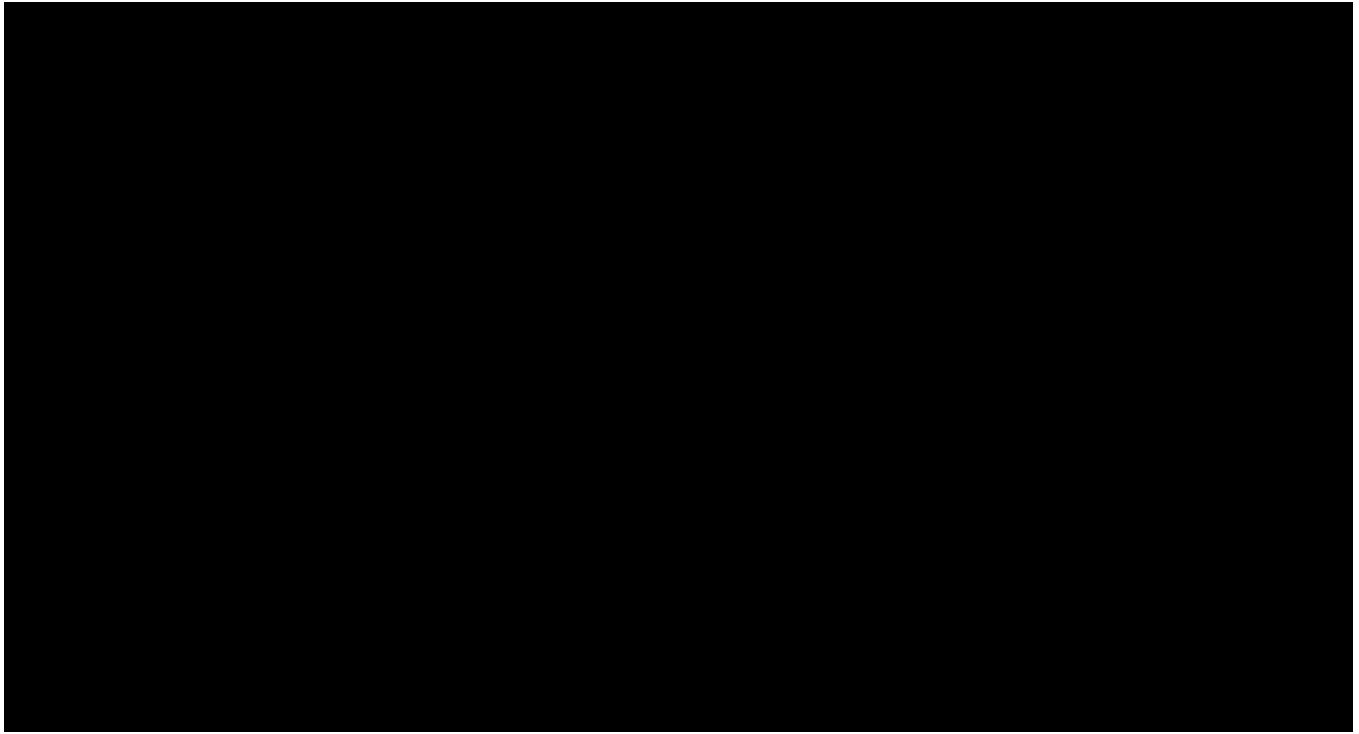
45. The Ni–Si phase diagram indicates that NiSi is a “line compound” with a very small tolerance for deviations from the exact stoichiometry. However, analyzing e.g. the profiles shown in Ex. A1 at 9-10 reveals that the ratio $X_{\text{Ni}}/X_{\text{Si}}$ is close to 1 only in two very narrow regions, one at $z = 20$ nm, the other one at $z = 25$ nm. In all other regions, significant deviations from the required stoichiometry are displayed in the plot. This explains why Complainant has neither identified a “nickel silicide surface at a silicon interface” nor an “opposite surface” of the silicide.



47. Complainant fails to provide an objective, reproducible criterion for the position of the Si interface that follows from these profiles.

48. If there were no uncertainty and no fluctuations related to the measurement technique, an objective, reproducible criterion could be the location where a profile changes curvature, e.g. from positive to negative curvature. Mathematically, that corresponds to a location where the second derivative of the profile goes through zero. However, such a criterion is not specified in the '235 patent. Moreover, such a criterion cannot even be applied to the profiles provided by Complainant. These profiles $W_i[z]$ were obtained by XEDS-TEM, and they demonstrate the weakness of this technique: significant fluctuations. If the same profiles were measured along a line at a different location of the same specimen, the result would be significantly different. This also explains why the profiles of Ex. 19 at 8, 9-10 look so different, although they should look the same. They show random fluctuations which may be based on limitations of the measurement technique or real fluctuations of compositions in the specimen.

In any event, such *irregular* profiles do not lend themselves to determine the position of surfaces/interfaces according to any agreeable definition.



51. In my opinion, Complainant's failure to identify the Si interface reflects the lack of certainty imposed by the irregularity of the profiles and the absence of an objective criterion for the location of "the Si interface." As a result, in my opinion, the term is indefinite.

2. Methodologies for Locating an Interface

52. A POSA would understand there are multiple methodologies for locating an interface.

53. For example, a paper entitled "Silicide-phase evolution and platinum redistribution during silicidation of $\text{Ni}_{0.95}\text{Pt}_{0.05}/\text{Si}(100)$ specimens" by Praneet Adusumilli, David N. Seidman, and Conal E. Murray ("Adusumilli") (Ex. 17) explores the Si/silicide interface using Atomic Probe Tomography (APT). Ex. 17 at 1. Using APT, the authors obtained highly accurate topographical 3D maps of *arbitrarily* chosen Si reference levels, e.g. $X_{\text{Si}} = 0.75$, a "75 at. % Si

isoconcentration surface.” Ex. 17 at 5. I have reproduced the topographical APT image from the paper below:

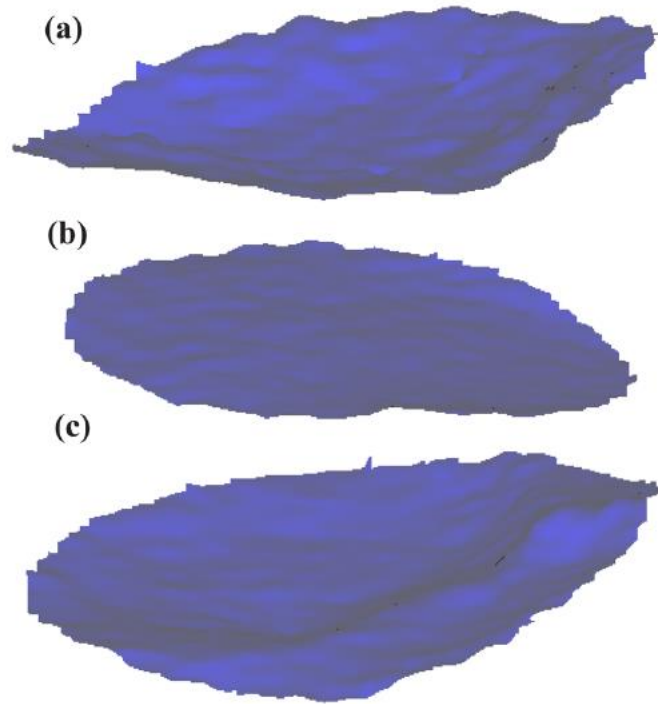


FIG. 7. Reconstruction of the silicide/Si(100) interface based on a 75 at. % Si isoconcentration surface for 10 nm thick ($\text{Ni}_{0.95}\text{Pt}_{0.05}$) film on Si(100) samples subjected to RTA at: (a) 320 °C, 5 s; (b) 420 °C, 5 s; and (c) 420 °C, 30 min. The root-mean-square chemical roughnesses of the interfaces are 2.1, 0.6, and 1.2 nm, respectively.

Ex. 17, Fig. 7.

54. Using this such Si isoconcentration surfaces as a *reference* (not a definition of a “Si interface” of any sort), Adusumilli obtained “proxigrams” – *three-dimensional* composition profiles that take into account the *local* shortest distance to the isoconcentration surface for each evaluated atom, thus the variation of the location of the isoconcentration surface. In this way, Adusumilli obtains *true* composition profiles $X_i[z]$ that are not impacted by lack of translation symmetry, whereas those in Ex. 19 at 8, 9-10 clearly are impacted.

55. The 3D (proxigram) profiles of Si, Ni, and Pt (exaggerated for visibility) are smooth and – compared to Ex. 19 at 8, 9-10 – exhibit only minimal fluctuations. Here, it can

make sense to define a unique location as one where graph curvature (second derivative) goes through zero. It makes even more sense noting that this position is the *same* for all three graphs, $X_{\text{Si}}[z]$, $X_{\text{Ni}}[z]$, and $X_{\text{Pt}}[z]$. In Fig 6, the final state after annealing, this position is indicated by a vertical dashed line. It intersects the Si profile at $X_{\text{Si}} = 0.70$, a somewhat lower level than the reference level of $X_{\text{Si}} = 0.74$ that had been chosen in this case to obtain the 3D profiles (proxigrams).

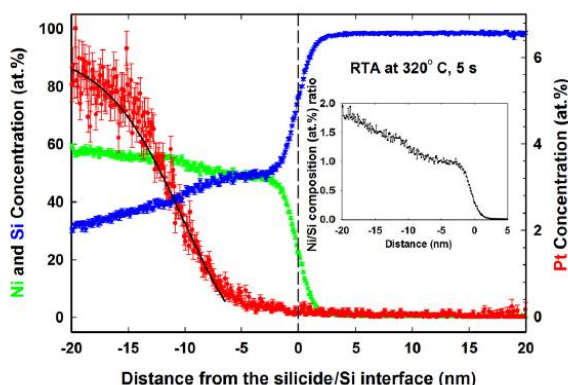


FIG. 4. Proximity histogram displaying the concentrations of Ni (green, $\blacktriangle$), Si (blue, $\blacktriangledown$), and Pt (red, $\bullet$) versus film depth for a 10 nm $\text{Ni}_{0.95}\text{Pt}_{0.05}/\text{Si}(100)$ film subjected to RTA at 320°C for 5 s. Concentrations were calculated for a series of 75 at. % Si isoconcentration surfaces moving into the thin film. On the abscissa zero represents the nickel-silicide/Si(100) heterophase interface. The black solid-line is a half-Gaussian fit to the Pt concentration profile. The inset diagram shows the Ni-to-Si concentration ratio (atomic percent) versus film depth.

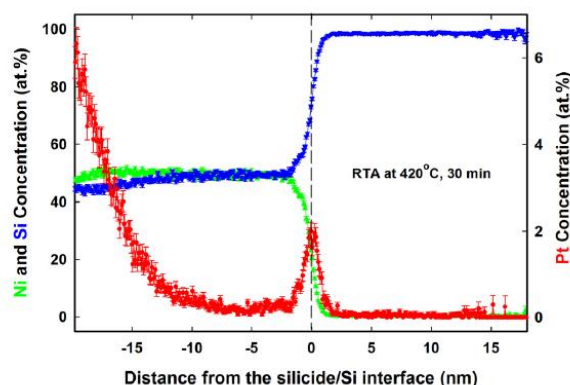


FIG. 6. Proximity histogram displaying the concentrations of Ni (green, $\blacktriangle$), Si (blue, $\blacktriangledown$), and Pt (red, $\bullet$) versus film depth for a 10 nm $\text{Ni}_{0.95}\text{Pt}_{0.05}/\text{Si}(100)$ thin film subjected to RTA at 420°C for 30 min. Concentrations were calculated for a series of 74 at. % Si isoconcentration surfaces moving into the thin film. Zero on the abscissa represents the nickel-silicide/Si(100) interface.

Ex. 17, Figs. 4, 6.

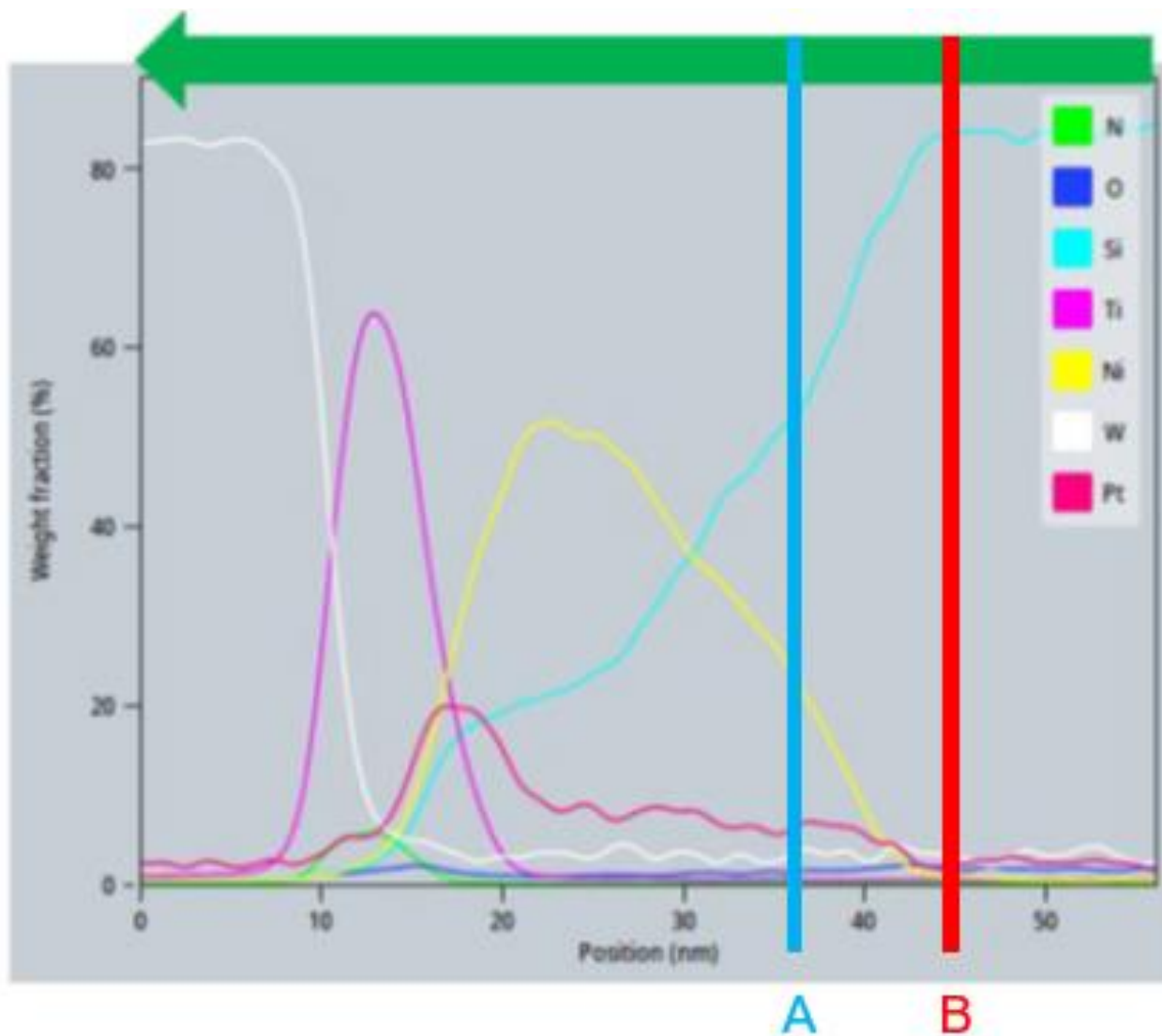
56. A different approach for locating the position of the “silicide surface” and the “silicon interface” emerges from the ability of TEM (transmission electron microscopy) not only to produce images and spectra, but also electron diffraction patterns from selected areas. These electron diffraction patterns form in the back-focal plane of the first lens after the specimen, called the objective lens. To switch the microscope from imaging the image plane of the objective lens to imaging the back focal plane of the objective lens onto the viewing screen, all that is needed is to adjust the focal length of the lens below, which is called the intermediate lens. Every transmission microscope can do this switch by the push of a button. If the selected area from

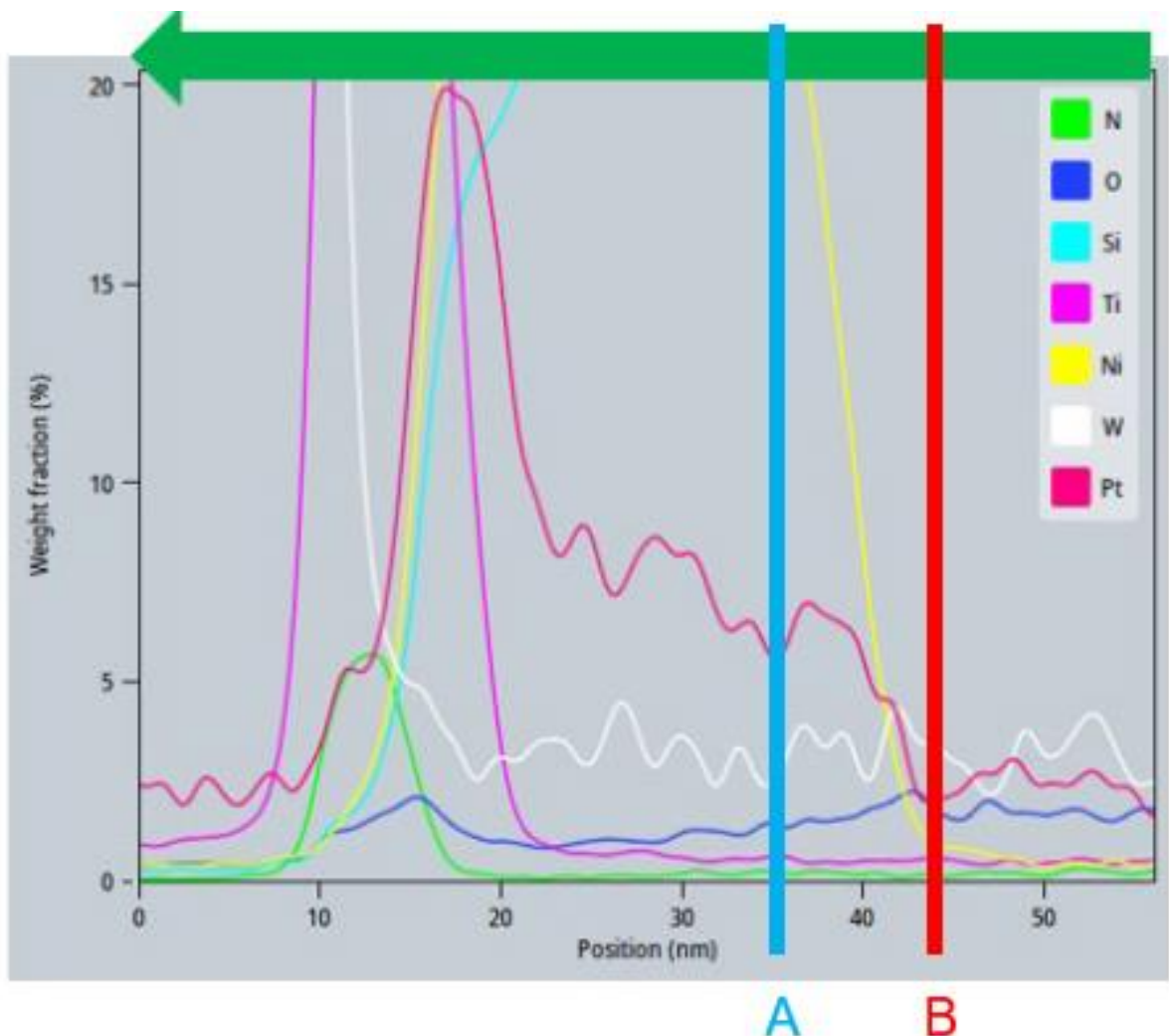
which are diffraction pattern is to be obtained were entirely in a region of Si (single-) crystal structure, a regular pattern of spots would appear on the screen. If, on the other hand, the selected area were entirely in a region of silicides, a different pattern of spots would appear. These spots will be in different locations than the spots that represent the crystal structure of Si. Therefore, by moving the selected area to different regions of the specimen, it can be determined whether they belong to the crystal structure of Si or the crystal structure(s) of silicides.

57. Moreover, it is possible to insert a circular *aperture* into the back-focal plane of the objective lens and position it onto one of the spots that belong to the crystal structure of Si, not to any crystal structure of silicides. Switching back to imaging (by pushing the “D” button again), a so-called dark-field image will be obtained. This image will show bright regions only where the specimen has the (single-) crystal structure of Si. Dark regions, in contrast, represent regions where the crystal structure is different. There will be a borderline between bright and dark regions. This borderline will mark the “silicide surface.” For several reasons, this approach to determining the position of the silicide surface is more reliable than inferring the position from composition profiles: The transition between crystal structures is intrinsically abrupt and the dark-field method provides spatial resolution that is close to the resolution limit of the transmission electron microscope, usually better than 1 nm. Inferring the position of the “silicide surface” from composition lineprofiles, in contrast, suffers from worse spatial resolution and other significant uncertainties, as described before. For example, as mentioned above, Ex. A1 at 9-10 reveals that the ratio $X_{\text{Ni}}/X_{\text{Si}}$ is close to 1 only in two very narrow regions, one at $z = 20$ nm, the other one at $z = 25$ nm, in eclatant contradiction with the anticipated lateral extension of the silicide NiSi in this specimen.

58. A person of skill in the art would understand that applying these different methodologies can – *will* – lead to different results.

59. This is significant because, as shown by the annotated below from Complainant's infringement contentions, depending where the interface is located can lead to different determinations about the amount of platinum:





Ex. 19, Complainant's Infringement Contentions, Ex. A1 at 10 (annotated).

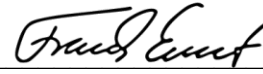
60. At location A, the mass fraction of Pt is $W_{Pt} = 0.07$ (7 wt%). At location B, in contrast, $W_{Pt} = 0.03$ (3 wt%). At both locations, the Pt mass fraction falls *outside* of the claimed range of $0.08 \leq W_{Pt} \leq 0.12$ ((8..12)wt%) required at the “silicon interface.” Since the mass fractions at these locations do not fall within the scope of claim 1, Complainant is obviously using some unknown, undisclosed methodology for picking a different location of the “silicon interface.”

61. In view of the above, in my opinion, the term “from a nickel silicide surface at a Si interface to an opposite surface of the nickel silicide” lacks reasonable certainty to a POSA.

PUBLIC VERSION

62. I declare under penalty of perjury that the foregoing is true and correct.

Dated: August 25, 2026

A handwritten signature in black ink, appearing to read "Frank Ernst", is positioned above a horizontal line.

Dr. Frank Ernst